

SCAMMON BAY, AK, USA

WMO: 703054

Lat: 61.845N

Lon: 165.571W

Elev: 14

StdP: 14.69

Time Zone: -9.00 (NAK)

Period: 05-19

WBAN: 00108

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -19.8	(c) -15.4	(d) -25.9	(e) 1.3	(f) -19.9	(g) -19.9	(h) 1.8	(i) -14.4	(j) 34.8	(k) 16.6	(l) 32.4	(m) 20.7	(n) 2.5	(o) 100	(p) 0.874

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.5	65.9	58.7	63.0	56.8	60.1	55.5	59.6	64.3	57.6	61.6	56.0	59.4	9.6	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
57.4	70.2	61.5	55.4	65.3	59.1	54.1	62.3	58.2	26.2	64.2	24.9	61.7	23.8	59.4	72.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
31.8	27.1	23.9	-18.4	73.8	4.8	5.4	-21.9	77.6	-24.7	80.8	-27.4	83.8	-30.9	87.7
			-18.2	63.0	4.7	4.4	-21.6	66.1	-24.4	68.7	-27.0	71.1	-30.5	74.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	29.7	6.8	12.5	9.3	21.3	36.5	48.3	52.9	52.4	46.2	34.8	20.1	13.9		
	DBStd	19.43	15.87	14.16	11.68	9.99	7.21	6.23	4.90	4.38	5.14	6.74	10.56	11.30		
	HDD50	7695	1338	1050	1262	861	421	101	22	24	129	470	898	1119		
	HDD65	12891	1803	1470	1727	1311	882	500	376	392	563	935	1348	1584		
	CDD50	282	0	0	0	0	4	52	113	97	16	0	0	0		
	CDD65	2	0	0	0	0	0	0	2	0	0	0	0	0		
	CDH74	8	0	0	0	0	0	0	8	0	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	9.4	7.3	10.3	9.3	9.3	9.1	10.5	10.0	9.7	9.8	9.9	9.7	7.7		
	PrecAvg	28.1	1.9	1.9	1.3	1.2	1.5	2.1	3.0	3.9	3.8	2.8	2.6	2.1		
(15) Precipitation	PrecMax	38.3	4.3	4.6	4.2	2.4	3.2	3.2	4.6	5.8	6.7	5.6	5.1	4.4		
	PrecMin	22.6	0.4	0.6	0.2	0.3	0.2	0.9	1.8	1.3	0.6	0.8	0.5	0.5		
	PrecStd	3.7	0.9	1.0	0.9	0.5	0.7	0.6	0.8	1.0	1.1	1.0	1.2	1.1		
	PrecStd	3.7	0.9	1.0	0.9	0.5	0.7	0.6	0.8	1.0	1.1	1.0	1.2	1.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.5	36.3	35.9	42.9	61.4	69.1	73.3	69.4	60.8	50.9	42.4	39.2		
		MCWB	36.6	34.5	34.3	39.2	53.3	58.8	63.9	60.4	54.8	47.3	40.1	38.3		
	2%	DB	35.4	34.4	33.7	38.5	54.2	63.7	66.4	64.2	56.9	48.5	38.4	35.5		
		MCWB	33.8	33.2	32.5	36.1	48.7	56.2	59.7	57.5	53.3	45.9	36.9	34.3		
	5%	DB	33.5	33.6	31.0	36.2	49.8	59.9	63.3	61.4	54.7	46.5	36.3	33.6		
		MCWB	32.5	32.5	30.2	34.3	46.1	54.5	57.4	57.0	51.7	44.5	35.0	32.7		
	10%	DB	29.8	31.5	27.7	34.2	46.0	57.0	60.6	58.5	53.3	44.7	33.9	30.4		
		MCWB	29.0	30.8	26.6	32.8	42.9	52.6	56.0	55.3	51.0	42.7	32.8	29.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.9	35.0	34.5	39.6	52.9	60.1	65.9	61.2	56.5	48.4	40.9	38.3		
		MCDB	38.7	35.5	35.2	42.4	58.6	66.5	72.4	66.1	57.8	49.6	42.0	38.8		
	2%	WB	34.3	33.4	32.5	36.3	49.1	57.2	60.7	59.0	54.2	46.5	37.2	34.4		
		MCDB	35.5	33.9	33.4	38.1	52.8	62.4	65.5	63.0	55.8	47.9	38.2	34.9		
	5%	WB	32.3	32.2	30.2	34.3	45.9	55.1	58.0	57.3	52.3	44.8	35.1	32.7		
		MCDB	33.1	33.1	30.9	35.7	49.6	59.1	62.3	60.7	54.0	46.5	36.1	33.4		
	10%	WB	28.8	30.4	26.6	32.8	43.2	53.1	56.1	55.7	50.7	42.8	32.7	30.0		
		MCDB	29.5	31.2	27.6	34.0	46.0	56.6	59.9	58.6	52.5	44.6	33.7	30.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.3	11.2	11.7	10.1	8.7	9.3	8.5	7.7	7.0	6.4	8.8	9.7		
		MCDBR	10.4	10.4	11.3	10.5	15.1	14.2	13.9	12.0	9.8	7.1	8.5	9.1		
		MCWBR	9.8	9.8	10.1	8.8	10.2	8.8	8.6	7.3	6.6	6.3	8.6	8.8		
		MCWBR	10.4	10.4	9.9	10.1	14.5	13.2	13.0	10.4	8.4	7.1	8.1	9.3		
	5% WB	MCWBR	10.1	10.1	9.5	8.7	10.4	8.8	8.5	6.9	6.5	6.5	8.4	9.1		
		MCWBR	10.1	10.1	9.5	8.7	10.4	8.8	8.5	6.9	6.5	6.5	8.4	9.1		
(40) Clear-Sky Solar Irradiance	taub		0.217	0.282	0.288	0.355	0.375	0.373	0.382	0.400	0.340	0.302	0.250	0.210		
	taud		2.033	2.121	2.192	2.042	2.201	2.393	2.406	2.301	2.465	2.353	2.022	1.880		
	Ebn at Noon		153	213	261	260	266	271	264	246	245	208	128	83		
	Edh at Noon		11	22	30	43	40	34	32	33	23	18	12	8		
(44) All-Sky Solar Radiation	RadAvg		97	328	887	1426	1614	1643	1323	1052	693	346	108	45		
	RadStd		13	38	97	105	161	158	93	145	61	29	17	5		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page